Brandenburgische Technische Universität Cottbus - Senftenberg

Internship Module

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Internship Report

Introduction:

My name is Afaq Ur Rehman Khan, and I am currently pursuing a Master of Science in Artificial Intelligence at Brandenburg University of Technology (BTU) Cottbus-Senftenberg, Germany. As part of my academic journey, I have had the opportunity to work as a software developer in a working student position at Fraunhofer FOKUS, one of the leading research centers in Germany. This report provides a comprehensive overview of my internship experience at Fraunhofer, detailing the various responsibilities and projects I undertook during my tenure. The internship was an integral component of my academic program, offering practical exposure and hands-on experience in the field of mobile and web development while allowing me to apply theoretical knowledge acquired through my coursework in artificial intelligence.

Through this role, I aimed to enhance my technical skills, gain valuable industry insights, and contribute to the ongoing projects at Fraunhofer. This report will elaborate on my objectives, the tasks I handled, the challenges I faced, and the learning outcomes achieved. The experience not only enriched my professional capabilities but also significantly contributed to my personal growth and readiness for future career endeavors in the research and development sectors.

Company Background:

The Fraunhofer-Gesellschaft was founded in Munich on March 26, 1949, as part of a program to reorganize and expand Germany's research infrastructure. The organization takes its name from Joseph von Fraunhofer (1787 - 1826), the successful Munich researcher, inventor, and entrepreneur. In its early years, the main function of this non-profit organization was predominantly administrative: to raise funds through government bodies, donations, and association members for distribution to research projects of relevance to industry.

Initial activities primarily focused on industry in Bavaria. This being the early post-war period, there was a particular need for research in the fields of mining, the iron and steel industry, and mechanical engineering. Over the decades, Fraunhofer has grown into Europe's largest application-oriented research organization, with 72 institutes and research units spread across Germany. The organization's research efforts now cover a broad spectrum of scientific

disciplines and applications, aiming to drive innovation and foster technological advancements that benefit industry and society.

Fraunhofer FOKUS, where I interned, is one of the institutes under the Fraunhofer umbrella. It focuses on research and development in the areas of smart cities, digital networking, and communication systems. The institute collaborates with partners from industry, academia, and the public sector to develop solutions that address current and future challenges in these domains. My role at Fraunhofer FOKUS was centered around the development of mobile and web applications, particularly in the context of media streaming and the Metaverse, providing a unique opportunity to contribute to cutting-edge research and innovation.

Internship Objectives:

During my tenure as a working student at Fraunhofer, I was primarily involved in the development of the Media Streaming project, which aimed to collect Common Media data from Media Players and send it to servers to ensure the best playback quality for the players and users. This role encompassed both technical skill development and the practical application of theoretical knowledge gained from my Master's program in Artificial Intelligence. These objectives were designed to align with my academic goals and career aspirations, providing a well-rounded experience in the field of mobile and web development.

1. **Enhance Technical Proficiency:**

Frontend Development:

One of the key objectives was to enhance my skills in web development, specifically through learning and applying ReactJS. This included understanding how to create dynamic and responsive user interfaces, and leveraging libraries to adhere to project requirements. Mastering ReactJS required understanding its core concepts such as component-based architecture, state management, and the use of props to pass data between components.

Mobile Application Development:

To complement my mobile application development work, I learned Java and Kotlin to handle mobile application operations. I focused on writing code to gather data from Mobile Applications and send it to the server for better playback quality. Additionally, I worked on

Bluetooth Low Energy (BLE) to gather data from a real bike sensor and send it to the server for the Metaverse project. These apps facilitated the retrieval, manipulation, and storage of data, ensuring that the mobile app could respond effectively to user inputs and actions. Understanding the intricacies of BLE and its integration into mobile applications was a significant technical challenge and learning opportunity.

2. Apply Theoretical Knowledge in a Practical Setting:

The internship provided an opportunity to apply the theoretical concepts learned during my MS in Artificial Intelligence program to real-world projects. This included understanding the principles of software development, data management, and system integration within a practical context. For instance, concepts such as machine learning, data mining, and algorithm optimization were applied in the context of improving media playback quality and user experience in the Media Streaming project.

3. Contribute to the Streaming and Metaverse Project:

My role in the projects was critical, as it aimed to develop and optimize operational processes of Media and Metaverse Applications. Contributing to this project involved developing key components of the mobile and web application, ensuring that it met the functional and aesthetic requirements set by the project stakeholders.

4. Develop Collaborative and Professional Skills:

- Working within a professional environment like Fraunhofer allowed me to develop important soft skills such as teamwork, communication, and project management. Collaborating with experienced developers and participating in team meetings and discussions helped me understand the dynamics of working in a professional setting.
- Regular interactions with my supervisors and peers provided invaluable feedback, helping me refine my development practices and align with industry standards. The ability to effectively communicate technical details and project progress was crucial in ensuring smooth collaboration and project success.

5. Achieve Measurable Outcomes:

Setting clear, measurable goals such as completing specific tasks, developing particular features of the web application, and meeting project deadlines was an important objective. These outcomes provided tangible evidence of my progress and contributions throughout the

internship. For example, successfully developing a fully functional media player with integrated analytics and monitoring features demonstrated my technical capabilities and commitment to the project's success.

6. Prepare for Future Career Opportunities:

Ultimately, the internship was designed to prepare me for future career opportunities in the field of mobile and web development and artificial intelligence. By gaining hands-on experience with modern technologies and methodologies, I aimed to build a strong foundation that would enhance my employability and readiness for future roles. The exposure to real-world projects and the opportunity to work with cutting-edge technologies provided a significant boost to my confidence and professional preparedness.

These objectives guided my activities and focus during the internship, ensuring that I derived maximum benefit from the experience while contributing effectively to the projects at Fraunhofer. The structured approach to achieving these objectives allowed me to systematically enhance my skills, apply theoretical knowledge, and gain valuable professional experience.

Job Responsibilities:

During my tenure as a working student at Fraunhofer, my responsibilities spanned various aspects of mobile and web development, project management, and collaboration. These responsibilities allowed me to apply my academic knowledge in a real-world setting while gaining valuable professional experience.

Frontend Development with ReactJS:

- **Learning and Application**: I dedicated time to learning ReactJS, a widely-used JavaScript library for building user interfaces. This included understanding its core concepts, such as components, state, and props. Additionally, I familiarized myself with various libraries and tools that enhance ReactJS development, such as Redux for state management, React Router for navigation, and Material-UI for styling.
- **Web Page Development**: Using ReactJS, I developed several web pages for the Media Streaming project. I utilized different libraries to ensure that the styling and layout adhered to the project requirements, creating responsive and interactive UI components that provided a seamless user experience. The development process involved close collaboration with UI/UX designers to ensure that the final product met both functional and aesthetic standards.

Research & Development on Media Streaming Apps:

- **App Development**: To support the Mobile App, I learned Java and Kotlin to handle mobile app operations. I used ExoPlayer to play the videos and did customization to retrieve metrics from the player and send them to the server to ensure better playback quality for the users. I designed and implemented Android apps that facilitated data interactions between the mobile app and backend, ensuring efficient data retrieval and manipulation
- **Integration**: I integrated APIs with the Mobile App, enabling the Android app to respond effectively to user inputs and actions. This involved designing and implementing RESTful APIs, ensuring secure and efficient communication between the frontend and backend systems.

Research & Development on Metaverse Apps:

- **App Development**: I acquired skills in working on Bluetooth Low Energy (BLE), focusing on how to gather data in Android apps from a bike sensor and use it to simulate the bike in a metaverse application. This involved understanding the BLE protocol, developing custom services and characteristics, and ensuring reliable data transmission.
- **Data Handling**: This involved structuring, querying, and managing data within the mobile app to ensure efficient data storage and retrieval processes, supporting mobile app operations. Effective data handling was crucial for ensuring real-time responsiveness and accuracy in the metaverse simulation..

Research & Development on Maven Package Registry:

- **Learning Maven Package Registry:** I acquired skills in managing and deploying an AAR or SDK via Maven Package Registry. It helped the other developers and team members to install the packages whenever needed as a dependency. Understanding the intricacies of Maven's dependency management and repository structure was essential for successful implementation.

Version Controlling with Git:

- **Version Control:** I implemented version control using Git, managing code repositories, and ensuring proper versioning of the codebase.
- **Feature-Branching:** I used feature branching for Isolation of Changes, Parallel Development, Better Code Review Process, Enhanced Collaboration, and Simplified Rollbacks.

Summary of Responsibilities:

- **ReactJS Frontend Development**: Developed responsive and interactive web pages using ReactJS and different component libraries.
- **Research & Development of Media Streaming Applications**: Designed and implemented mobile apps to support CMCD operations, ensuring efficient data retrieval and manipulation.
- **Research & Development of Media Streaming Applications**: Designed and implemented mobile apps to transmit Bluetooth Low Energy (BLE) data and perform operations.
- **Research & Development on Maven Package Registry**: Implemented Maven Package Registry to help developers and team members install dependencies efficiently.
- **Version Controlling with Git**: Implemented Git for version control and used feature branching for better management of the repository.

These responsibilities provided a comprehensive experience in mobile and web development and project management, allowing me to develop both technical and collaborative skills essential for my future career in artificial intelligence and software development. The hands-on experience with various technologies and tools significantly enhanced my technical proficiency and preparedness for future roles.

Collaborative Projects:

During my internship at Fraunhofer, I had the privilege of working on several collaborative projects that required close coordination with team members, including two full stack developers, one manager, and one senior developer. The primary project I contributed to was the Media Streaming project, which aimed to transmit metrics from video players and enhance the playback quality for the users.

Media Streaming Project:

Team Collaboration:

This project required extensive collaboration among team members. As a mobile app developer, I worked closely with the UI/UX designers to ensure that the user interfaces I developed were not only functional but also aesthetically pleasing and user-friendly. Regular meetings and design review sessions facilitated this collaboration, allowing for real-time

feedback and iterative improvements. The collaborative environment fostered a culture of continuous learning and improvement, enabling us to deliver a high-quality product.

API Integration:

I collaborated with other full stack developers to integrate the backend services with the mobile app. This involved designing and implementing RESTful APIs using Retrofit and Volley and ensuring seamless data flow between the client and server. Effective API integration was crucial for enabling real-time data transmission and ensuring a seamless user experience.

Version Control:

We used Git for version control, with each team member responsible for managing their branches and merging code changes through pull requests. This process was overseen by the scrum master and involved thorough code reviews to maintain code quality and consistency. The use of feature branching allowed us to isolate changes and ensure that each feature was fully developed and tested before integration into the main codebase.

User Testing:

The QA engineers played a crucial role in this aspect, conducting thorough testing of the interfaces to identify any usability issues or bugs. Collaborating with them ensured that the final product met the high standards expected by the company. The iterative testing and feedback loop allowed us to identify and address issues early in the development process, ensuring a robust and reliable product.

Bug Fixing:

Identified bugs were tracked in Jira Boards, and as a team, we prioritized and addressed these issues in subsequent sprints. This collaborative approach ensured that the product remained robust and reliable. The use of agile methodologies and tools like Jira facilitated efficient project management and ensured that we met our project deadlines and milestones.

Metaverse Project:

Data Collection and Simulation: For the Metaverse project, I collaborated with other developers to integrate BLE data collection into the mobile app. This required close coordination with hardware engineers to ensure accurate data transmission from the bike sensor to the app. The collected data was then used to simulate the bike's movements in a virtual environment, providing users with an immersive experience.

Integration and Testing: The integration of BLE data into the Metaverse application posed several challenges, including ensuring real-time data transmission and synchronization with the virtual environment. Collaborating with other team members allowed us to effectively address these challenges and optimize the app's performance.

Challenges Faced:

During my internship, I encountered several challenges that tested my problem-solving abilities and adaptability. These challenges provided valuable learning experiences and helped me grow both technically and professionally.

Technical Challenges:

Learning New Technologies:

One of the initial challenges was learning ReactJS, Maven Package Registry, and Bluetooth Low Energy (BLE), which were new to me. The steep learning curve required me to dedicate extra time to study and practice these technologies to effectively contribute to the project. Overcoming this challenge involved leveraging online resources, tutorials, and guidance from my supervisors and peers.

Complex Integrations:

Integrating the web frontend and mobile applications with the backend services and ensuring smooth communication between them posed a technical challenge. This required a deep understanding of both frontend and backend technologies, as well as effective debugging skills to resolve integration issues. Tackling this challenge involved extensive testing, debugging, and collaboration with other developers to identify and address potential issues.

Task Prioritization:

Managing multiple tasks and prioritizing them based on project needs was another challenge. Balancing immediate bug fixes with long-term feature development required careful planning and time management.

Collaborative Challenges:

Communication Barriers:

Working with a diverse team meant overcoming communication barriers, particularly when coordinating with team members from different technical backgrounds and expertise areas. Ensuring clear and effective communication was crucial to avoid misunderstandings and delays.

Achievements:

Despite the challenges, I achieved several milestones during my internship at Fraunhofer, which contributed to my professional growth and the success of the projects I worked on during my tenure..

Technical Achievements:

Frontend Development:

Successfully developed and implemented multiple web pages using ReactJS and different components libraries, ensuring that they were responsive and aligned with the project's design specifications.

Mobile App Development:

Designed and implemented robust Mobile apps using Java & Kotlin enabling efficient communication between the frontend and backend.

Version Controlling:

Set up and maintained repositories using Git, Using feature branching significantly improved the efficiency and reliability of the project's deployment workflow.

Task Completion:

Successfully completed all assigned tasks, including bug fixes, feature development, and integration tasks, contributing to the overall progress and success of the projects.

Learning Outcomes:

The internship at Fraunhofer provided a wealth of learning opportunities, significantly enhancing my technical, collaborative, and project management skills.

Technical Learning Outcomes:

Proficiency in ReactJS:

Gained a thorough understanding of ReactJS, enabling me to develop complex, responsive web applications. This included mastering concepts like state management, props, and component lifecycle.

Mobile App Development Skills:

Acquired skills in mobile app development using Java and Kotlin, including integrating RESTful APIs and working with ExoPlayer, Maven Package Registry and Bluetooth Low Energy (BLE).

Version Controlling:

Learned to use Git for version control and using feature branching to ensure Isolation of Changes, Parallel Development and other benefits

Teamwork and Communication:

Enhanced my ability to work effectively within a team, improving my communication and collaboration skills. This was critical for coordinating tasks, resolving conflicts, and ensuring project success.

Task Prioritization and Time Management:

Improved my ability to prioritize tasks and manage my time effectively, ensuring that I met deadlines and contributed to the project's progress.

Conclusion:

My internship at Fraunhofer was an invaluable experience that provided me with significant professional and personal growth opportunities. Working on the Media Streaming and Metaverse projects allowed me to apply and enhance my technical skills in mobile and

web development, while also developing crucial project management and collaborative skills.

Overall Experience:

- The internship provided a comprehensive learning experience, bridging the gap between theoretical knowledge and practical application. It prepared me for future challenges in the field of artificial intelligence and software development.
- The collaborative environment at Fraunhofer, along with the support from my supervisors and team members, played a crucial role in my development. Their guidance and feedback were instrumental in overcoming challenges and achieving project milestones.

Future Aspirations:

- The skills and experiences gained during this internship have significantly contributed to my academic and career aspirations. I feel more prepared and confident to tackle complex projects and work effectively within a professional team setting.
- Moving forward, I aim to leverage the skills and knowledge gained at Fraunhofer to pursue a successful career in artificial intelligence and mobile and web development, contributing to innovative projects and solutions in the tech industry.

In conclusion, my time at Fraunhofer was a transformative experience that provided me with the tools and confidence to excel in my future career endeavors. I am grateful for the opportunity and look forward to applying what I have learned in my ongoing professional journey.